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BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Profit-Aware Telecom Churn & Retention Targeting

Introduction to Business Analytics · Capstone Topic 7 · Assoc. Prof. Nguyen Binh Minh

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Outline

I. Business problem
Decision, data, risk versus value

II. Model and decision rule
Calibrated churn model, expected value, optimiser

III. Results and robustness
Same-budget comparison, budget size, sensitivity

IV. Recommendation and demo
Randomised pilot, live dashboard

The Decision: Who Gets a Retention Offer?

Given a limited retention budget, which customers should receive an offer when both churn risk and expected value count?

- 26.5% of customers left; churn puts about $468,468 of 12-month margin at risk
- Today the fixed budget goes to the customers most likely to leave
- But every offer costs money, and not every risky customer is worth saving

Churn risk
p, from the model

×

Save rate
q, assumed per offer

×

Customer value
V, 12-month margin

−

Cost
offer + contact

→

Offer if EV > 0
EV = p × q × V − cost

Save rate = share of would-be leavers an offer keeps. Money in USD of the fictional sample company.

Data and Validation Design

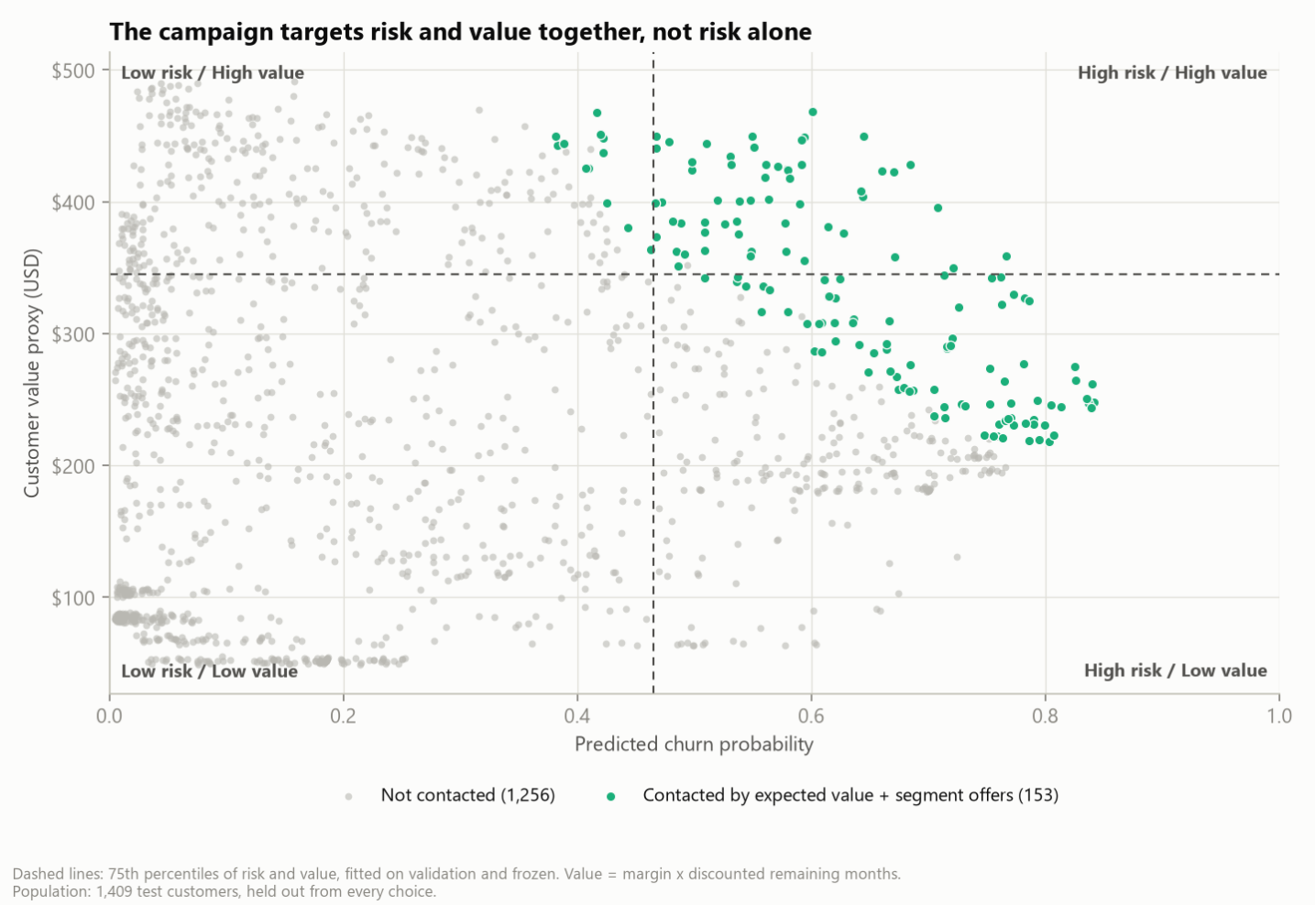
7,043
customers, IBM Telco sample (fictional US operator)

26.5%
churned in the sample window

60 / 20 / 20
train / validation / test, stratified

Item	Decision
Unit of analysis	One customer
Target	Churn: left during the sample window (yes / no)
Decision	Offer or not, and which offer, within the budget
Data quality	0 duplicates; 11 blank TotalCharges (tenure 0) set to 0
Leakage control	Churn reasons and IBM score excluded; TotalCharges dropped
Validation	4,225 / 1,409 / 1,409; test used once
Fairness	Gender and senior status audited, never model inputs

Key Insight: Churn Risk Is Not Customer Value



Dashed lines: 75th percentiles of risk and value, fitted on validation and frozen. Value = margin x discounted remaining months.
Population: 1,409 test customers, held out from every choice.

46%

of value at risk sits with high-risk, low-value customers, each worth little

57

test customers are high-risk and high-value; risk-only targeting reaches 12%

- Churn: month-to-month 43% vs two-year 3%
- New customers 48%, fibre 42%, electronic check 46%
- Associations, not causes

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Churn Model: Simple, Calibrated, Good Enough

Model (test split)	PR-AUC	ROC-AUC	Brier	ECE
Prior-only baseline	0.265	0.500	0.195	0.022
Logistic regression	**0.620**	**0.830**	**0.142**	**0.019**
Random forest	0.621	0.832	0.144	0.045
XGBoost	0.613	0.826	0.145	0.044

1,409 test customers held out from every choice; calibrated models; the baseline predicts the churn rate for everyone

×3.7
odds: month-to-month vs two-year contract

×2.9
odds: fibre vs no internet

×0.69
odds per extra 12 months of tenure

0.83
ROC-AUC; PR-AUC 0.62 vs 0.27 baseline

0.019
calibration error: probabilities fit to multiply by money

- Logistic regression chosen by a rule fixed in advance: tree models did not rank better
- Drivers are associations, not causes

From Probability to Offer: Expected Value

$$EV = p(\text{churn}) \times \text{save rate} \times \text{customer value} - (\text{offer cost} + \text{contact cost})$$

Customer value

Monthly bill × 30% margin × discounted 12 months

Horizon ×0.75 for new and ×1.25 for loyal customers

10% annual discount rate

Offer catalogue (assumed)

$35 credit for moving to a 1-year contract: save rate 30%

6 months free TechSupport + OnlineSecurity: 15%

One-time $20 bill credit: 20%

Plus $5 contact cost per customer

Optimiser

Integer program maximises total EV within the budget

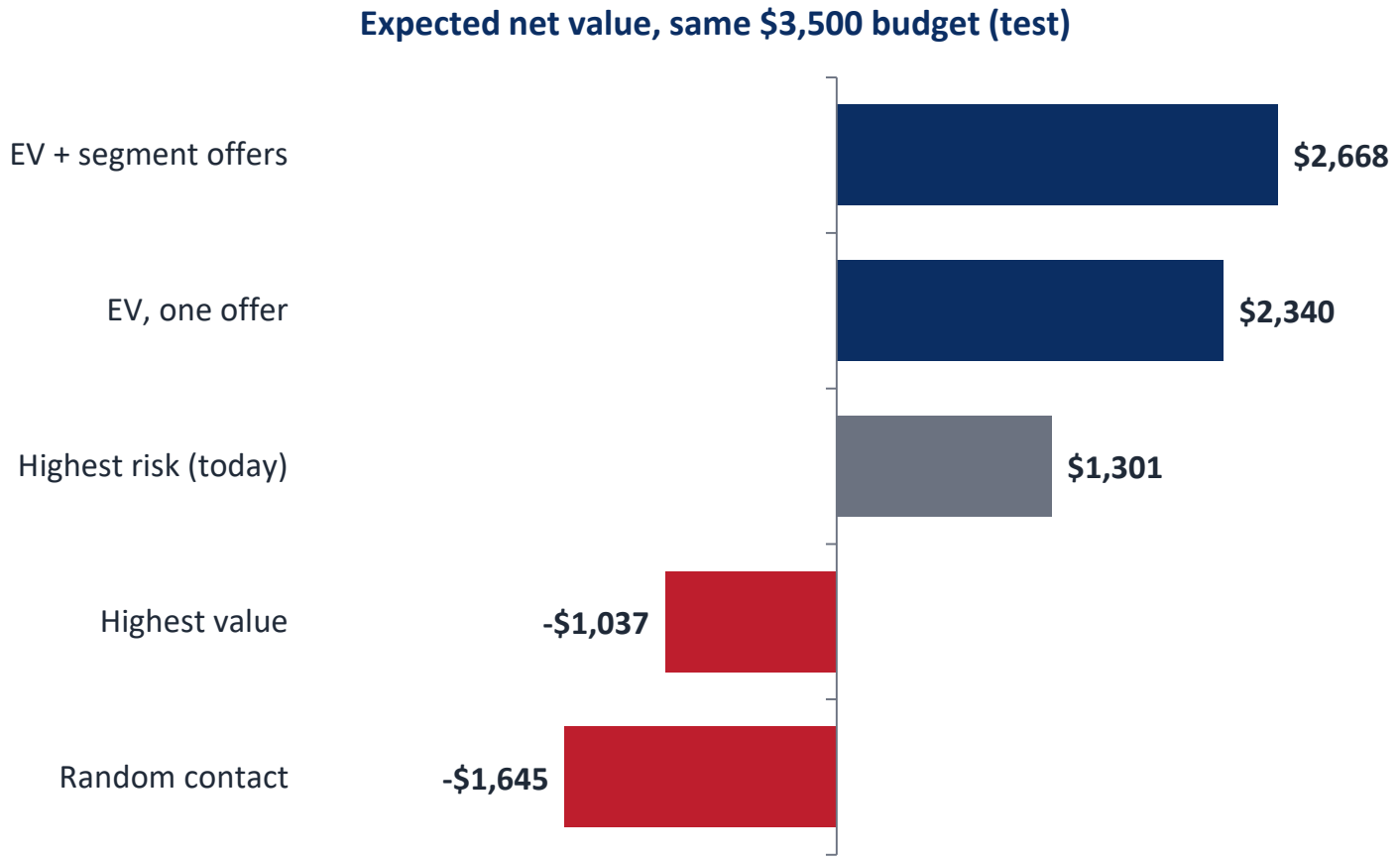
One offer per customer; eligibility rules

At most 30% of customers contacted

EV ≤ 0: never contacted

Save rates and costs are visible assumptions, not measurements: the data contain no randomised campaign, so no causal uplift is claimed.

Same Budget, $1,368 More Expected Value



+$1,368

vs the highest-risk rule; ROI 76% vs 37%

+$1,077

valued with actual churn; 95% interval $229 to $2,002

- Risk-only is more precise (71% vs 60%) but spends on customers worth little
- Targets: 100% fibre, 97% month-to-month, median bill $89

Expected values under the stated assumptions (USD, fictional sample company), not realised profit.

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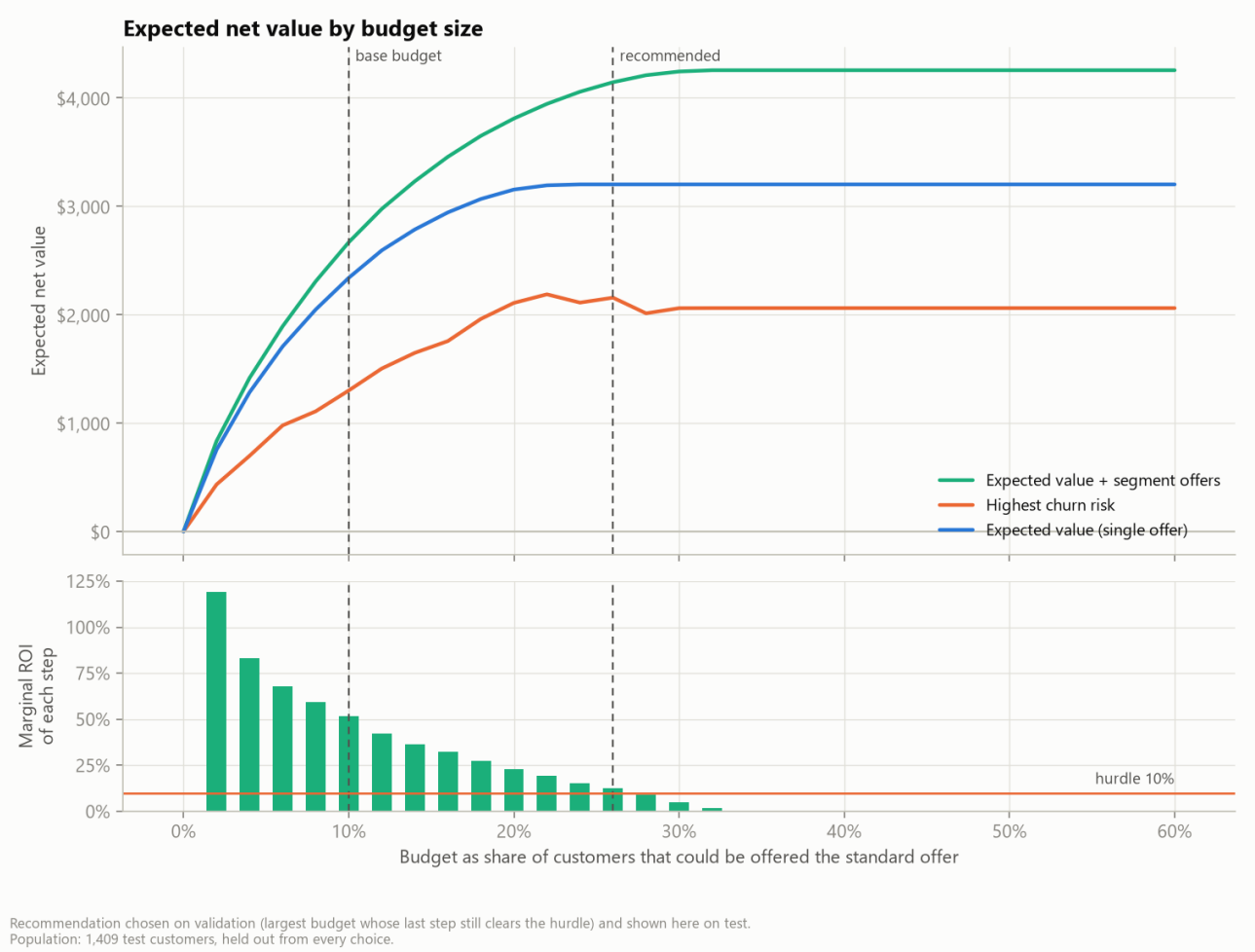
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How Much to Spend: The Profit Curve



Recommendation chosen on validation (largest budget whose last step still clears the hurdle) and shown here on test.
Population: 1,409 test customers, held out from every choice.

26%

budget share chosen on validation: the last step must still return ≥ 10%

$4,145

expected net value on test: 324 offers for $9,150 (ROI 45%)

- About $45,737 for the 7,043-customer base ($6.49 per customer)
- Positive EV runs out at 371 test customers ($11,318); the hurdle stops earlier, where extra steps barely pay

How Sure Are We?

99.96%
of 5,000 draws: EV targeting beats risk-only

13.8%
break-even save rate of the standard offer for the recommended list (assumed 20%)

17%
chance the recommended list loses money

46%
of churners left for price or a competitor: what an offer can address

Who to target is robust; how much an offer saves is not, so measure it before scaling

Biggest sources of uncertainty
Value horizon (months), save rate of the standard offer and contribution margin

Fairness check
Seniors contacted 3.3× as often (1.7× after adjusting for risk); age and gender are not model inputs; every wave is reviewed

Recommendation: A Randomised Pilot Before Scaling

Set-up
4 weeks
Retention, Legal/DPO, IT

→

Pilot wave
13 weeks
Call centre, Retention

→

Read-out
2 weeks
Retention analytics

→

Scale or stop
4 weeks
Head of Retention, Finance

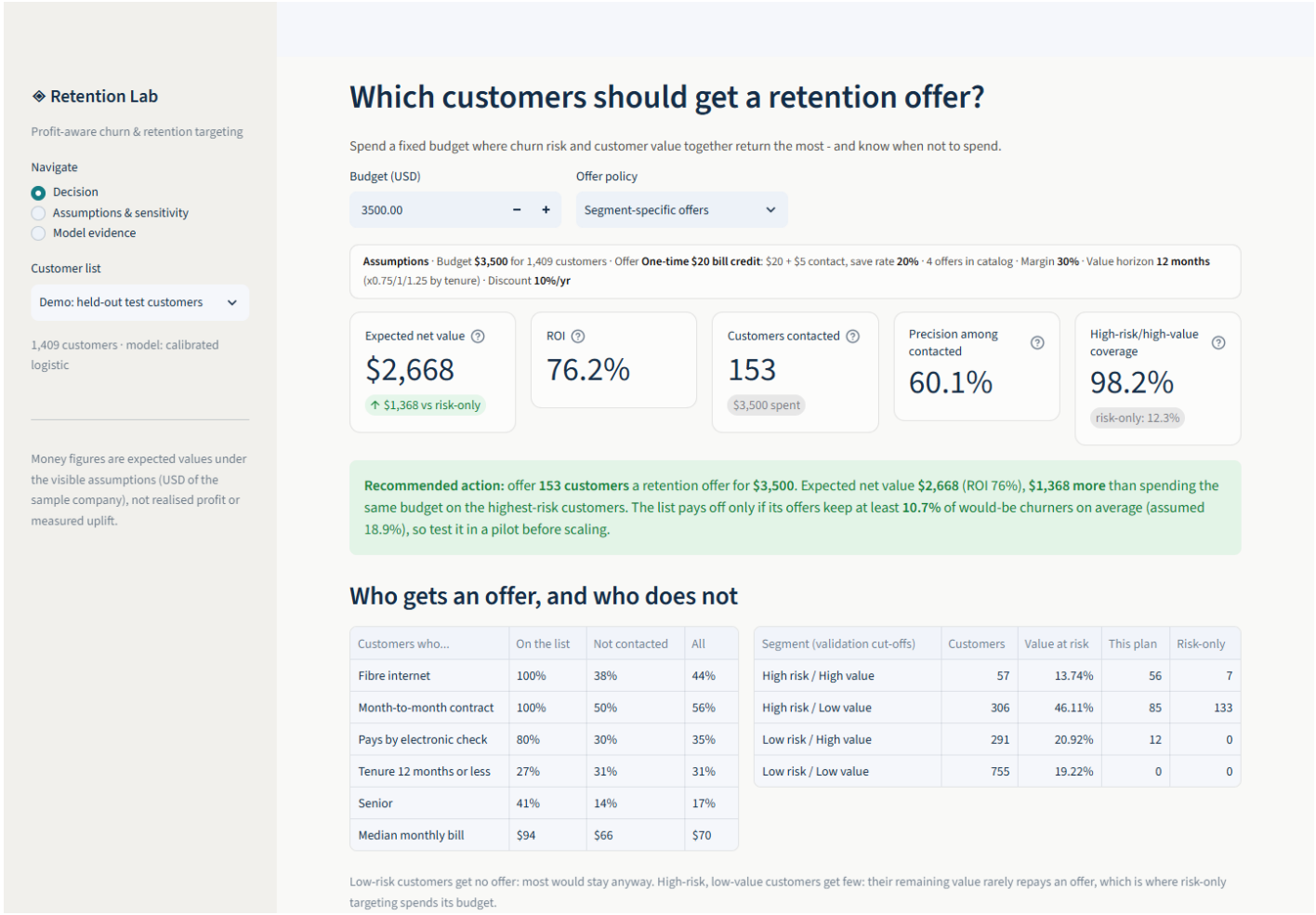
1,620
customers: 810 offered, 810 held out at random

$22,875
cost of the offered half; expected net value $10,363

≥ 15.6%
measured save rate of the offer mix needed to go (assumed 22.6%); P(go) = 97% if it holds

Adopt expected-value targeting with segment offers, launched as a randomised pilot; scale only if the measured save rate clears break-even

Live Demo: Retention Lab Dashboard



Decision

Five KPIs, recommendation, who gets an offer, action list and export

Assumptions & sensitivity

Change save rates and costs; break-even, live chance of a loss, stress test

Model evidence

Performance, points scorecard, SHAP challenger, score an uploaded customer file

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THANK YOU !

Questions and discussion

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